Data Landscape — Open Standards for Modern Data Architecture

Curated by Entropy Data · www.data-landscape.com

DEFINITION — HOW DATA IS DESCRIBED

CONTRACTS (6)

- ADOPT — ODCS — BITOL @ LF
- ADOPT — OpenAPI — LF
- ADOPT — AsyncAPI — LF
- ADOPT — GraphQL — LF
- SITUATIONAL — gRPC — CNCF
- SITUATIONAL — OData — OASIS

DATA PRODUCTS (4)

- ADOPT — ODPS — BITOL @ LF
- ASSESS — DPDS — ODM
- ASSESS — ODPS — LF
- ASSESS — DPROD — OMG

SCHEMA (7)

- ADOPT — XML Schema — W3C
- ADOPT — JSON Schema — IETF
- ADOPT — SQL DDL — ISO/IEC
- ADOPT — AVRO Schema — ASF
- ADOPT — Protobuf — GOOGLE
- ASSESS — LinkML — LINKML
- ASSESS — Table Schema — FRICTIONLESS

SEMANTICS (9)

- SITUATIONAL — RDF/OWL — W3C
- SITUATIONAL — DCAT — W3C
- SITUATIONAL — SKOS — W3C
- SITUATIONAL — SHACL — W3C
- SITUATIONAL — JSON-LD — W3C
- ASSESS — Ossie Semantic Model — OSSIE @ ASF
- ASSESS — Ossie Ontology — OSSIE @ ASF
- ASSESS — schema.org — W3C
- ASSESS — ShEx — W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS (8)

- ADOPT — CSV — IETF
- ADOPT — JSON — IETF
- ADOPT — XML — W3C
- ADOPT — YAML — YAML.ORG
- ADOPT — PARQUET — ASF
- ADOPT — AVRO — ASF
- SITUATIONAL — ORC — ASF
- ASSESS — Lance — LANCE

OPEN TABLE FORMATS (4)

- ADOPT — Iceberg — ASF
- SITUATIONAL — Delta — LF
- ASSESS — Hudi — ASF
- ASSESS — Lance — LANCE

STORAGE SYSTEMS (2)

- ADOPT — S3 — AWS
- CAUTION — HDFS — ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY (5)

- ADOPT — JDBC — JCP
- ADOPT — ODBC — ISO/IEC
- ADOPT — ADBC — ASF
- ASSESS — Quack — DUCKDB LABS
- CAUTION — XMLA — MICROSOFT

INTERCONNECTION (3)

- ADOPT — HTTP — IETF
- SITUATIONAL — Delta Sharing — LF
- CAUTION — FTP / SFTP — IETF

MESSAGING (5)

- ADOPT — Kafka — ASF
- ADOPT — CloudEvents — CNCF
- SITUATIONAL — MQTT — OASIS
- SITUATIONAL — AMQP — OASIS
- CAUTION — JMS — JAKARTA EE

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT (2)

- ADOPT — DataFrame API — DATA-APIS
- ADOPT — Apache Arrow — ASF

PROCESSING (7)

- ADOPT — Spark — ASF
- ADOPT — Pandas — NUMFOCUS
- ADOPT — SQL DML — ISO/IEC
- SITUATIONAL — dbt — DBT LABS
- ASSESS — Beam — ASF
- CAUTION — Ibis — IBIS
- CAUTION — XSLT — W3C

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS (5)

- ADOPT — Iceberg Catalog — ASF
- ADOPT — Schema Registry — CONFLUENT
- SITUATIONAL — Unity Catalog — LF
- ASSESS — DuckLake — DUCKDB LABS
- CAUTION — Hive Metastore — ASF

LINEAGE (2)

- ADOPT — OpenLineage — LF
- ASSESS — PROV — W3C

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY (5)

- ADOPT — SQL — ISO/IEC
- SITUATIONAL — Substrait — LF
- SITUATIONAL — SPARQL — W3C
- SITUATIONAL — GQL — ISO/IEC
- CAUTION — MDX — MICROSOFT

DATA QUALITY (3)

- SITUATIONAL — Great Expectations — GX LABS
- SITUATIONAL — dbt tests — DBT LABS
- SITUATIONAL — SodaCL — SODA

OBSERVABILITY (2)

- ADOPT — OpenTelemetry — CNCF
- ASSESS — OORS — BITOL @ LF

POLICIES (2)

- ADOPT — OPA — CNCF
- ASSESS — ODRL — W3C

AI INTERFACES (2)

- ADOPT — MCP — LF
- SITUATIONAL — A2A — LF